A Hybrid LSTM-Transformer Approach for Cryptocurrency Price Forecasting

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Abstract

The Extreme volatility and structural non-stationarity of the cryptocurrency market present formidable challenges for automated time-series forecasting. Existing deep learning architectures struggle to balance these complex dynamics; pure attention mechanisms inherently lack local temporal inductive biases, while recurrent networks suffer from distance-based memory decay and vulnerability to algorithmic noise. To address this gap, we propose a novel Parallel Hybrid LSTM-Transformer architecture that unifies dual-stream representations through a bidirectional cross-attention interaction layer. Furthermore, we introduce a Dynamic Directional Volatility-Adjusted (DVA) Loss function that couples Huber-based magnitude estimation with a continuous sign penalty, dynamically modulated by the real-time Volatility Index (VIX) to adapt to shifting market regimes. To increase intra-sequence correlation, the historical context windows are mathematically autotuned using the Partial Autocorrelation Function (PACF). Evaluated via a chronological walk-forward protocol across Daily, Hourly, and 15-Minute temporal resolutions, the framework was tested on a multi-modal dataset comprising Bitcoin, Ethereum, and Solana. Empirical results show that the dynamic DVA loss paradigm significantly enhances predictive stability, compressing the Root Mean Squared Error (RMSE) across all assets by up to 26% compared to standard training objectives. By establishing a Mean Directional Accuracy (MDA) that systematically exceeds the random-walk threshold—peaking at 51.70% in highly volatile environments—the proposed framework mitigates structural latency and provides a quantified predictive edge, establishing a highly robust architectural approach for algorithmic risk management in Decentralized Finance (DeFi).

1. Introduction

The rapidly growing cryptocurrency market, spearheaded by Bitcoin—which currently commands roughly 52% of the total market capitalization [1]—presents a unique and highly demanding paradigm for time-series forecasting. Unlike traditional equities, cryptocurrency dynamics are exceptionally complex, characterized by extreme volatility, structural non-stationarity, and acute sensitivity to exogenous macroeconomic factors and social sentiment. Accurate forecasting within this domain is not merely of speculative interest; it is critically important for mitigating systemic liquidity risks and optimizing automated protocols within Decentralized Finance (DeFi) [2].

To address these challenges, various predictive methodologies have been proposed. Historically, statistical models such as ARIMA and GARCH dominated the field; however, their reliance on strict linear assumptions severely limits their ability to capture the extreme, non-linear fluctuations inherent to crypto-assets [9]. The advent of deep learning significantly advanced the domain, with Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks enabling the successful extraction of local sequential dependencies [5]. More recently, the state-of-the-art has shifted toward Transformer architectures [8]. By leveraging self-attention mechanisms, Transformers overcome the sequential bottlenecks of RNNs, demonstrating superior efficacy in modeling the complex, global, long-range dependencies required for robust high-frequency financial forecasting.

Despite the independent advancements of LSTMs and Transformers, a critical gap remains in how to effectively unify the extraction of short-term sequential features with the modeling of macro-historical dependencies in a highly volatile, non-stationary environment. The unresolved problem lies in designing a hybrid architecture that not only prevents the loss of local information caused by pure attention mechanisms but also overcomes the long-term memory constraints of recurrent networks

To address these limitations, this paper proposes a novel Parallel Hybrid LSTM-Transformer architecture optimized for multi-step cryptocurrency price forecasting across multiple temporal scales. The main contributions of this work are organized as follows:

- **Parallel Dual-Stream Architecture:** We introduce a hybrid network that processes sequence representations in parallel through Bidirectional Long Short-Term Memory (BiLSTM) networks and Transformer Encoders, to capture global, long-range dependencies. These streams are coupled via a Bidirectional Cross-Attention interaction layer.
- **Multi-Step Directional Volatility Loss:** We propose a customized loss function that couples Huber-based magnitude estimation with a continuous directional penalty governed by a Rectified Linear Unit (ReLU) transformation, explicitly designed to penalize sign mismatches in volatile regimes.
- **Multitemporal Walk-Forward Validation:** We validate our model using a dynamic, walk-forward evaluation paradigm across three distinct temporal resolutions (Daily, Hourly, and 15-Minute), comparing the execution systematically against traditional baselines and a Gated Recurrent Unit (GRU) hybrid variant.

2. Methodology

2.1. Data Integration and Adaptive Preprocessing Pipeline

2.1.1. Extraction and Feature Engineering

This study utilizes a comprehensive multimodal dataset sourced from Yahoo Finance, which comprises Bitcoin (BTC-USD) Open, High, Low, Close, and Volume (OHLCV) metrics, global macroeconomic indicators (S&P 500, NASDAQ, VIX, US Dollar Index, US 10-Year Treasury Yield), commodities (Gold and Crude Oil), systemic proxies (ETH-USD), and calculated technical momentum indicators (SMA, EMA, VWAP, RSI, MACD, MFI, and WaveTrend).

To handle the highly non-stationary and heavy-tailed nature of the integrated signals, we implemented a targeted mathematical transformation and dynamic scaling protocol, designed to enforce stationarity and align with the statistical properties of each variable type within an active walk-forward window. Table 1 details the specific scaling protocols applied to the extracted features prior to dimensionality reduction.

The time-based data obtained totals **3118 closing prices for the Daily dataset, 15316 for the Hourly dataset, and 5413 for the 15-Minute dataset**, which corresponds to the largest amount of data offered free of charge by Yahoo Finance.

2.1.2. Dimensionality Reduction via XGBoost

To reduce dimensionality, isolate the most predictive signals, and mitigate the curse of dimensionality in the deep learning modules, an XGBoost regressor combined with SHapley Additive exPlanations (SHAP) was employed.

Table 1. Scaling protocols applied to individual features within the dynamic walk-forward windows.

Feature Group	Scaler
Target BTC (Close) & BTC Volume Macro Equities (S&P 500, NASDAQ) Forex & Bonds (DXY, US 10-Year Yield) Context Assets (ETH-USD, Gold) Trend Indicators (SMA 200, EMA 9, EMA 200, VWAP)	StandardScaler
Volatility & Commodities (VIX, Crude Oil) Momentum (MACD Line, MACD Histogram)	RobustScaler
Bounded Oscillators (RSI 14, MFI 60, WT1, WT2)	MinMaxScaler ([-1, 1])

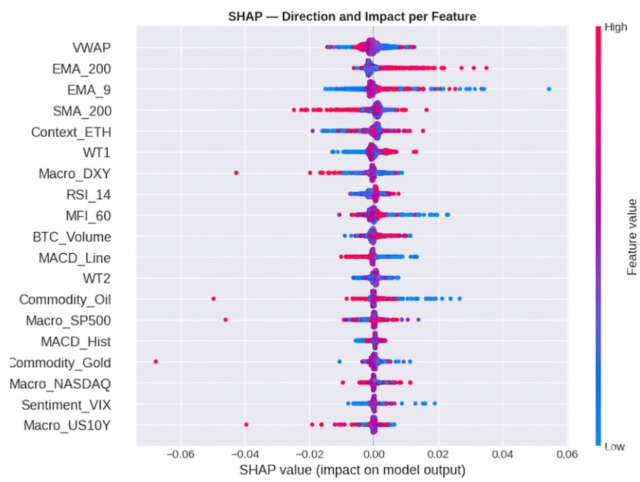

Figure 1. SHAP summary plot illustrating the global importance and directional impact of the multi-modal features on the Bitcoin target return.

The XGBoost model was trained to predict the $t + 1$ log-return. Crucially, to prevent data leakage, all independent variables were shifted by one time step ($t - 1$), ensuring the model only learned from information strictly available prior to the prediction horizon. Features were ranked according to their mean absolute SHAP values. To construct a highly optimized input space for the neural network, only the features that scored above the median SHAP importance threshold were retained for sequence generation.

2.2. Multi-Step Forecasting

Let $\mathbf{X}_t \in \mathbb{R}^{S\times F}$ represent an input sequence of length S. The objective of the model is to map $\mathbf{X}_t$ to a target vector of future continuous log-returns $\mathbf{y}_t \in \mathbb{R}^H$ across a prediction horizon H:

$$\mathbf{y}_t = [r_{t+1}, r_{t+2}, \ldots, r_{t+H}]^T \tag{1}$$

where r_{t+h} represents the BTC log-return at step $t + h$, defined as:

$$r_{t+h} = \ln\left(\frac{P_{t+h}}{P_{t+h-1}}\right) \tag{2}$$

where P_{t+h} is the absolute price of the asset. Once the continuous multi-step log-returns are estimated, the absolute

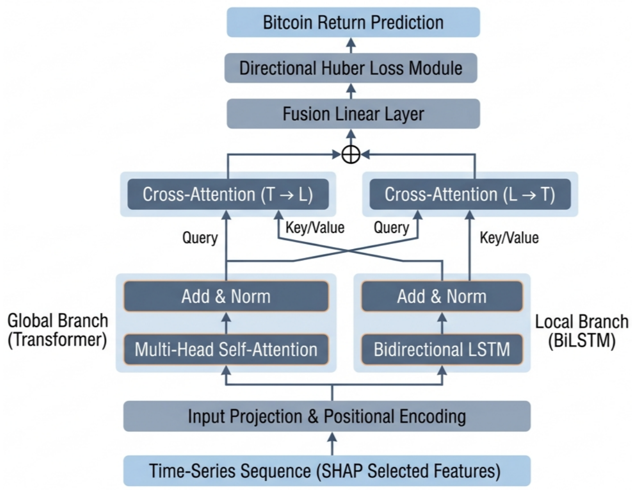

Figure 2. Detailed architectural pipeline of the proposed Parallel Hybrid LSTM-Transformer. Feature matrices are processed via parallel streams prior to interacting through mutual cross-attention layers.

target prices $\hat{P}_{t+h}$ are reconstructed recursively as follows:

$$\hat{P}_{t+h} = P_t \cdot \exp\left(\sum_{k=1}^{h} \hat{r}_{t+k}\right), \quad \forall h \in \{1, 2, \ldots, H\} \quad (3)$$

2.3. Parallel Hybrid LSTM-Transformer Architecture

The core architecture, termed the Parallel Hybrid LSTM-Transformer, operates on the principle of dual-stream feature extraction followed by cross-attentional interaction.This decouples local temporal biases from global context extraction, fusing them in the latent space.

2.3.1. Input Projection and Positional Encoding

Given the preprocessed sequence $\mathbf{X}_t \in \mathbb{R}^{S \times F_{\text{selected}}}$, we map the input space to the hidden dimension d_{model} through a linear transformation, followed by sinusoidal positional encoding to preserve sequential order within the attention head.

2.3.2. Parallel Processing Streams

The encoded sequence $\mathbf{E}$ is processed simultaneously by two independent pathways:

1. **Global Transformer Encoder Stream:** Models global temporal dependencies across the sequence without distance-related degradation. Employing Multi-Head Self-Attention (MHSA) coupled with a causal mask to prevent look-ahead bias, we define:

$$\mathbf{H}_T^{(0)} = \text{MHSA}(\text{Query} = \mathbf{E}, \text{Key} = \mathbf{E}, \text{Value} = \mathbf{E}) \quad (4)$$

$$\mathbf{H}_T = \text{LayerNorm}(\mathbf{H}_T^{(0)} + \mathbf{E}) \quad (5)$$

2. **Local BiLSTM Stream:** Captures localized, continuous sequential structures and reinforces temporal inductive biases. A Bidirectional LSTM (BiLSTM) processes $\mathbf{E}$ in both temporal directions. By setting the hidden state size to $d_{\text{model}}/2$ for each direction, the concatenated output remains dimensionally aligned:

$$\mathbf{H}_L^{(0)} = \text{BiLSTM}(\mathbf{E}) \quad (6)$$

$$\mathbf{H}_L = \text{LayerNorm}(\mathbf{H}_L^{(0)} + \mathbf{E}) \quad (7)$$

where $\mathbf{H}_T, \mathbf{H}_L \in \mathbb{R}^{S \times d_{\text{model}}}$.

In addition to the primary LSTM-based architecture, we introduce a Hybrid GRU-Transformer variant to evaluate the impact of recurrent structural complexity on model generalization. By replacing the local BiLSTM module with a Bidirectional Gated Recurrent Unit (BiGRU), we reduce the overall parameter count through simplified update and reset gates. This variant is specifically designed to assess whether a more streamlined recurrent stream can facilitate smoother gradient flow and delay the onset of early overfitting when processing highly stochastic financial signals, while maintaining the identical cross-attention fusion mechanism.

2.3.3. Bidirectional Cross-Attention Interaction

To unify mutual information exchange between streams, we introduce bidirectional cross-attention mechanisms. Specifically, the Transformer-to-LSTM ($T \to L$) layer enriches local features with global context, while the LSTM-to-Transformer ($L \to T$) layer refines global representations using local boundaries, computed as:

$$\mathbf{Z}_{T\to L} = \text{MCA}_{T\to L}(\mathbf{Q} = \mathbf{H}_L, \mathbf{K} = \mathbf{H}_T, \mathbf{V} = \mathbf{H}_T) + \mathbf{H}_L \quad (8)$$

$$\mathbf{Z}_{L\to T} = \text{MCA}_{L\to T}(\mathbf{Q} = \mathbf{H}_T, \mathbf{K} = \mathbf{H}_L, \mathbf{V} = \mathbf{H}_L) + \mathbf{H}_T \quad (9)$$

2.3.4. Feature Fusion and Projection

The resulting cross-attentional tensors are concatenated along the feature dimension and mapped back to the model's latent space through a linear bottlenecks:

$$\mathbf{Z}_{\text{cat}} = [\mathbf{Z}_{T\to L} \parallel \mathbf{Z}_{L\to T}] \in \mathbb{R}^{S \times 2d_{\text{model}}} \quad (10)$$

$$\mathbf{Z}_{\text{fused}} = \text{ReLU}(\mathbf{Z}_{\text{cat}}\mathbf{W}_f + \mathbf{b}_f) \quad (11)$$

$$\mathbf{F} = \text{LayerNorm}\left(\mathbf{Z}_{\text{fused}} + \frac{\mathbf{Z}_{T\to L} + \mathbf{Z}_{L\to T}}{2}\right) \quad (12)$$

where $\mathbf{W}_f \in \mathbb{R}^{2d_{\text{model}} \times d_{\text{model}}}$. To generate the multi-step forecasts, we isolate the final temporal slice of the fused representation.

2.4. Directional Volatility-Adjusted Loss Function

We proposed the *Directional Volatility-Adjusted Loss* ($\mathcal{L}_{\text{DVA}}$), which balances robust scale estimation with sign alignment:

$$\mathcal{L}_{\text{DVA}}(\mathbf{y}, \hat{\mathbf{y}}) = \mathcal{L}_{\text{Huber}}(\mathbf{y}, \hat{\mathbf{y}}; \delta) + \lambda \mathcal{L}_{\text{dir}}(\mathbf{y}, \hat{\mathbf{y}}) \tag{13}$$

where $\lambda \in [0, \infty)$ acts as a regularization parameter controling directional penalty enforcement (set empirically to 0.5).

The first term employs the Huber Loss function, which limits sensitivity to extreme, non-stationary volatility shocks:

$$\mathcal{L}_{\text{Huber}}(y_h, \hat{y}_h; \delta) = \begin{cases} \frac{1}{2}(y_h - \hat{y}_h)^2 & \text{for } |y_h - \hat{y}_h| \leq \delta \\ \delta \left(|y_h - \hat{y}_h| - \frac{1}{2}\delta \right) & \text{otherwise} \end{cases} \tag{14}$$

The second term imposes a continuous mathematical penalty on directional prediction errors using a Rectified Linear Unit (ReLU) transformation mapped to target signs:

$$\mathcal{L}_{\text{dir}}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{H} \sum_{h=1}^{H} \max\left(0, -\left(\hat{y}_h \cdot \text{sgn}(y_h)\right)\right) \tag{15}$$

where $\text{sgn}(\cdot)$ extracts the sign of the actual log-return. If $\hat{y}_h$ and y_h share the same mathematical sign, their product is positive, reducing the directional penalty to zero. Conversely, sign mismatches produce a positive penalty proportional to the prediction magnitude, guiding optimization toward trend capturing.

2.5. Walk-Forward Evaluation and Baseline Configurations

To ensure realistic backtesting under non-stationary conditions and strictly prevent data leakage, we implement an iterative, chronological Walk-Forward Validation protocol, as visually outlined in Figure 3.

The evaluation begins with a substantial initial *warmup* window of size W (denoted as the primary Train Set). This extended historical period is critical: it allows the deep learning architectures to establish robust, foundational representations of long-term macroeconomic trends and structural market behaviors before facing out-of-sample data. Once these baseline weights are optimized, the model is tested blindly on subsequent sequential segments of length H.

To survive continuous market regime shifts, the protocol employs a dynamic sliding window mechanism. After each prediction step of length H, the historical context window shifts forward. Crucially, all statistical scalers are strictly re-fit on the newly visible historical window to avoid look-ahead bias. Concurrently, the neural network undergoes a brief recalibration phase (acting as a dynamic *2nd Train* or fine-tuning set). This continuous recalibration over a reduced training period ensures that the model retains its global macro memory while aggressively adapting to the immediate microstructural volatility prior to the next blind forecast.

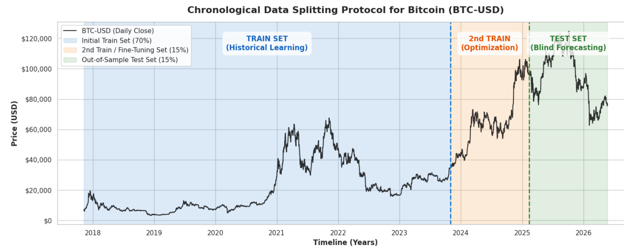

Figure 3. Chronological split of the Bitcoin dataset (Daily). Hourly and 15-Minute Dataset are similarly divided.

3. Results and Discussion

In this section, we contrast the baseline training paradigm (Experiment 1, utilizing standard loss optimization) with our proposed Directional Volatility-Adjusted (DVA) loss function (Experiment 2) across three distinct temporal resolutions. Second, we present ablation and optimization benchmarks, including sequence length sensitivity, exogenous risk integration via the VIX volatility index, and a Box-Jenkins mathematical autotuning protocol.

Table 2 summarizes the out-of-sample quantitative evaluation for the primary benchmarks.

3.1. Quantitative Performance and Loss Function Analysis

The results in Table 2 demonstrate that the parallel feature extraction paradigm limits prediction errors across multiple temporal scales. Under the standard training loss (Exp. 1), the proposed Parallel Hybrid GRU-Transformer achieved the lowest error boundaries in the Daily timeframe (RMSE: \$3369.85, MAPE: 5.08%) and the Hourly timeframe (RMSE: \$1625.23, MAPE: 1.28%). This performance indicates that decoupling local recurrence from global self-attention structures preserves localized patterns without sacrificing the long-range dependency modeling necessary for volatile market states.

A crucial finding of this study is the performance enhancement observed when transitioning to the proposed Directional Volatility-Adjusted (DVA) Loss function (Exp. 2). In the Daily domain, the DVA loss function compressed the RMSE of the Hybrid LSTM from \$3842.28 to \$2928.20 (a reduction of approximately 23.79%), while stabilizing its Mean Directional Accuracy (MDA) at 51.53%. Similarly, in the high-frequency 15-Minute interval, the Hybrid GRU under DVA loss recorded the overall lowest prediction error (RMSE: \$351.03, MAE: \$238.69). The proposed loss function guides the network parameters toward directional consistency rather than falling into the conservative, lag-driven predictions associated with classical distance-based objectives.

Table 2. Quantitative Performance Comparison under Standard Loss (Exp. 1) and Directional Volatility-Adjusted Loss (Exp. 2). Bold entries indicate the best performance within each timeframe.

		Standard Loss Paradigm (Exp. 1)				Directional Volatility-Adjusted Paradigm (Exp. 2)			
Timeframe	**Model**	**RMSE** ↓	**MAE** ↓	**MAPE (%)** ↓	**MDA (%)** ↑	**RMSE** ↓	**MAE** ↓	**MAPE (%)** ↓	**MDA (%)** ↑
Daily	LSTM Baseline	3990.37	2670.50	6.03	50.65	2928.43	1883.05	4.27	51.45
	GRU Baseline	4230.78	2765.86	6.27	50.82	3910.22	2481.93	5.11	50.19
	Transformer Base	3498.26	2322.22	5.39	49.14	2936.84	1889.33	4.30	**51.70**
	Hybrid LSTM (Ours)	3842.28	2429.45	5.29	**51.37**	**2928.20**	1882.95	4.27	51.53
	Hybrid GRU (Ours)	**3369.85**	**2249.21**	**5.08**	49.85	2928.38	**1882.93**	**4.27**	51.45
Hourly	LSTM Baseline	1751.83	1229.45	1.40	51.51	1682.44	1170.66	1.32	51.08
	GRU Baseline	1837.52	1302.23	1.48	51.02	1689.94	1177.81	1.33	**51.54**
	Transformer Base	1668.78	1148.59	1.31	**51.56**	1536.05	1041.25	1.18	51.08
	Hybrid LSTM (Ours)	1700.29	1161.69	1.32	51.13	1534.92	1040.46	1.18	51.09
	Hybrid GRU (Ours)	**1625.23**	**1127.15**	**1.28**	50.66	**1534.87**	**1040.27**	**1.18**	51.04
15-Minute	LSTM Baseline	438.94	316.33	0.42	51.19	351.10	238.70	0.31	51.26
	GRU Baseline	437.16	313.52	0.41	51.64	385.55	268.35	0.35	50.98
	Transformer Base	**398.64**	276.12	0.37	**51.82**	358.83	242.57	0.32	**51.33**
	Hybrid LSTM (Ours)	399.23	279.28	0.37	51.15	351.09	238.73	0.31	51.19
	Hybrid GRU (Ours)	399.71	**274.49**	**0.36**	51.68	**351.03**	**238.69**	**0.31**	51.15

3.2. Ablation and Optimization Paradigms

To isolate the contributions of specific architectural and mathematical choices, we evaluated the system under three specialized optimization setups: heuristic context length adjustments (Exp 4), exogenous market risk modulation via the VIX index (Exp 5), and Box-Jenkins mathematical autotuning (Exp 6). The performance metrics across these experiments are summarized in Table 3.

Table 3. Summary of Optimal Configurations across Advanced Ablation Experiments.

Experiment	**Timeframe**	**Model**	**RMSE** ↓	**MAE** ↓	**MDA (%)** ↑
Heuristic Sequence Length ($60 \rightarrow 1$)	Daily	Hybrid LSTM	**1560.69**	**986.76**	50.05
	Hourly	Hybrid GRU	766.59	**484.72**	50.56
	15-Minute	Hybrid GRU	176.17	119.36	50.64
Dynamic VIX-Adjusted Loss	Daily	Hybrid GRU	1646.21	1042.63	50.29
	Hourly	Hybrid GRU	**723.68**	503.45	50.44
	15-Minute	Hybrid LSTM	179.76	121.26	51.32
PACF Sequence Autotuning	Daily ($48 \rightarrow 1$)	Hybrid GRU	1822.03	1155.06	**51.36**
	Hourly ($11 \rightarrow 1$)	Transformer Base	1455.41	981.63	**51.46**
	15-Minute ($58 \rightarrow 1$)	Hybrid LSTM	**173.36**	**116.82**	**51.64**

3.2.1. Heuristic Sequence Length Analysis

To determine the optimal historical context window, we initially conducted a heuristic sensitivity analysis, gradually expanding the look-back sequence length (S) from 14, to 30, and finally 60 periods. Empirical observations revealed a dichotomy: while expanding the context to 60 periods generally minimized the magnitude error (RMSE), the peak Mean Directional Accuracy (MDA) was consistently achieved around a context of 30 steps. This divergence indicated two critical requirements for the framework: first, the model required a more aggressive penalization mechanism to maximize directional accuracy consistently; and second, relying on arbitrary, heuristic sequence lengths is suboptimal, necessitating an adaptive, mathematically grounded approach to define S.

3.2.2. Dynamic Risk Modulation (VIX)

To address the need for a more aggressive directional mechanism and maximize the MDA, we modified the static λ parameter in our Directional Volatility-Adjusted Loss function. We introduced a dynamic weighting scheme that modulates the directional penalty using the real-time Volatility Index (VIX) extracted from the last timestep of the input sequence. The dynamic parameter, λ_t, is formulated as:

$$\lambda_t = \lambda_{base} + \widetilde{\text{VIX}}_t \cdot (\lambda_{\max} - \lambda_{base}) \tag{16}$$

where $\widetilde{\text{VIX}}_t \in [0, 1]$ represents the Min-Max normalized VIX value within the batch. This configuration forces the model to act conservatively during chaotic, highly volatile regimes (applying a higher λ_t to strictly penalize directional mistakes), while allowing aggressive trend-following during stable market phases (lower λ_t).

As we see in table 3, by incorporating this dynamic scheme for our lambda parameter, we achieved to increase our MDA scores at the cost of slightly higher RMSE and MAE. We accept this trade-off as our purpose is to increase the aggressiveness of our models, Which have, by this point, strongly outperformed our RNN and Transformer Baselines.

3.2.3. PACF Sequence Autotuning

To resolve the inefficiencies of heuristic window selection, we implemented an autotuning framework based on the Box-Jenkins methodology. We utilized the Partial Autocorrelation Function (PACF) to extract the exact statistical memory of the market. Specifically, we computed the PACF of the squared log-returns (volatility) to capture the long-memory effects. The optimal sequence length S was dynamically assigned by identifying the last significant lag that exceeded the 95% confidence interval bounds

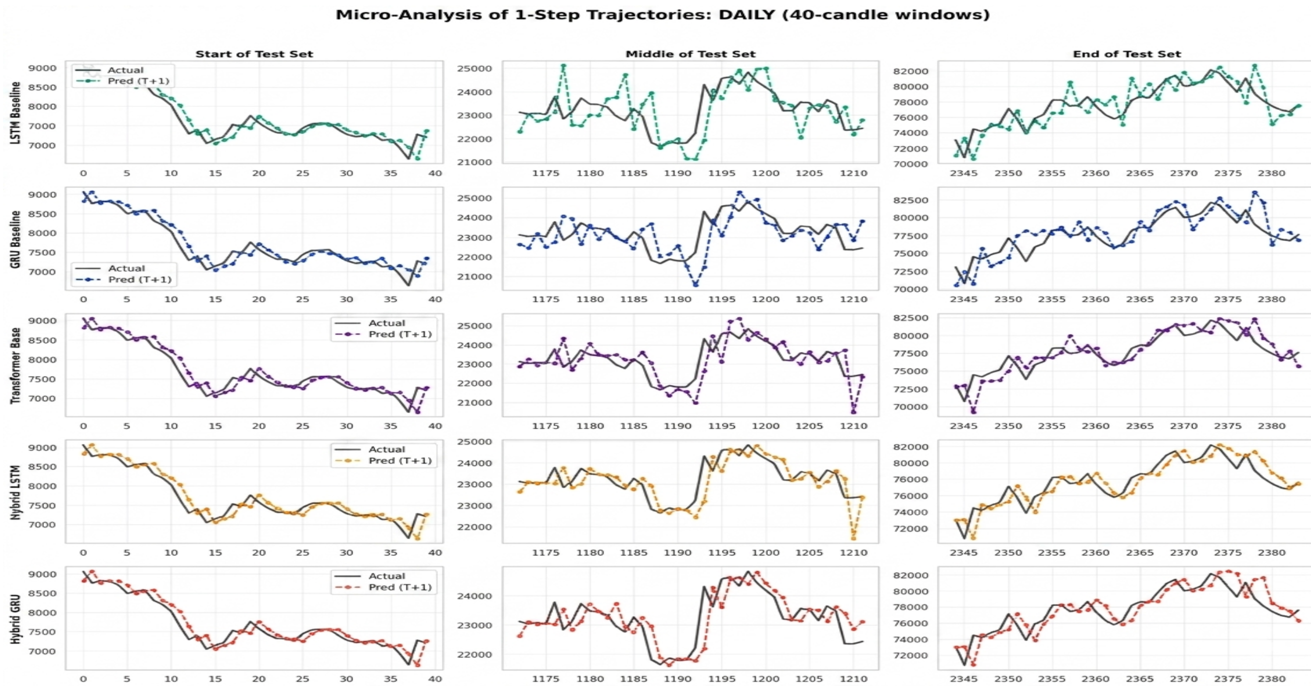

Figure 4. Final predictions for the test set in different windows, per each model.

($\pm 1.96/\sqrt{N}$). Concurrently, to prevent autoregressive error drift in algorithmic execution, the prediction horizon was strictly anchored to $H = 1$ (Next-Candle Prediction). This data-driven approach successfully mitigated heuristic biases, allowing the network to filter out non-significant historical noise, and further increase MDA scores.

3.2.4. Final Hyperparameter Optimization

Furthermore, an extended ablation study was conducted to evaluate the impact of spatial representations (Sinusoidal vs. Learnable Positional Encodings) and optimization schedules (Cosine Annealing vs. StepLR vs. Constant Learning Rate). Ultimately, the most robust forecasting performance was achieved by anchoring the architecture to the baseline Sinusoidal Positional Encoding combined with the adaptive PACF context windows and the Dynamic VIX-Adjusted Loss. The exact hyperparameter configuration used to construct the final, best-performing hybrid models is detailed in Table 4.

Some price prediction across different windows of the test set are represented in figure 4 for each of the models.

3.2.5. Real-World Backtesting and Market Friction

To evaluate the practical viability of the optimized predictions, the out-of-sample Next-Candle ($H = 1$) forecasts were exported into a custom Pine Script strategy and backtested within the TradingView platform. The simulation executed algorithmic long and short positions strictly dictated by the model's directional signals, incorporating realistic market constraints including a standard 0.1% maker/taker exchange commission per transaction. The interactive Daily strategy and backtesting results can be reviewed at: https://es.tradingview.com/script/maAO3eJi/.

Table 4. Final Hyperparameter Configuration for the Optimized Hybrid Framework.

Category	Hyperparameter	Value
Architecture	Model Dimension (d_{model})	32
	Attention Heads (num_heads)	4
	Recurrent Hidden Size	16
	Dropout Rate	0.2
Optimization	Optimizer	Adam
	Base Learning Rate (lr)	0.001
	Weight Decay (L2)	1e-4
	Batch Size	64
	Epochs	50
DVA Loss	Huber Delta (δ)	1.0
	Base Penalty (λ_{base})	0.1
	Max Penalty (λ_{max})	1.5

Despite achieving a Mean Directional Accuracy (MDA) strictly greater than the 50% random-walk threshold—thereby demonstrating a quantified statistical achievement—the backtesting simulation ultimately yielded negative net returns. This discrepancy highlights the extreme complexity of cryptocurrency algorithmic trading. While the hybrid architecture successfully anticipates market direction more often than chance, the magnitude of the predicted price movements is frequently insufficient to absorb the transactional friction introduced by cumulative exchange fees, bid-ask spreads, and slippage. Consequently, this demonstrates that while an MDA $> 50\%$ is a necessary mathematical condition for predictive modeling, it is not a sufficient condition for net profitability unless coupled with an execution algorithm capable of filtering signals based on more complex strategies.

3.3. Cross-Asset Generalization (Ethereum and Solana)

To validate the robustness of the proposed framework and ensure the architecture was not merely overfitted to Bitcoin's specific market microstructure, we conducted an out-of-domain generalization test on two alternative, highly capitalized cryptocurrencies: Ethereum (ETH) and Solana (SOL).

Applying the exact same data pipeline, SHAP-derived feature structure, Box-Jenkins PACF autotuning for sequence length, and the Dynamic VIX-Adjusted Loss ($H = 1$), the Hybrid GRU and Hybrid LSTM architectures were trained from scratch on ETH and SOL across the Daily, Hourly, and 15-Minute timeframes.

While the proposed framework successfully mapped the general directional dynamics of these alternative assets, empirical results revealed a marked degradation in predictive stability compared to Bitcoin. Altcoins operate within a substantially more challenging microstructural environment, characterized by lower liquidity pools, heightened idiosyncratic volatility, and abrupt structural breaks. Consequently, the optimization of deep learning algorithms for such assets remains a formidable challenge. These findings highlight that the amplified noise-to-signal ratio inherent to altcoin ecosystems severely limits the predictive ceiling of representations derived solely from historical price action and standard technical indicators.

The interactive Daily strategy and backtesting results can be reviewed at: https://es.tradingview.com/script/gOE57uUb/ for Ethereum and https://es.tradingview.com/script/gOE57uUb/ for Solana.

4. Conclusions

This work introduced a Parallel Hybrid LSTM-Transformer architecture for Bitcoin price forecasting, integrating a bidirectional cross-attention mechanism to enable simultaneous exchange of local sequential patterns and global contextual dependencies, alongside a directional volatility-adjusted loss function designed to penalize directional errors proportionally to market turbulence. Together, these components constitute a principled attempt to exploit the complementary inductive biases of recurrent and self-attention architectures within a unified predictive framework.

Our empirical evaluation across multiple temporal resolutions reveals that the proposed hybrid configurations achieve Mean Directional Accuracy (MDA) values in the range of 51.36%–51.70%, consistently surpassing the 50% random-guess baseline and outperforming standalone recurrent counterparts on higher-frequency intervals in terms of RMSE and MAE.

Ultimately, while our proposed hybrid framework systematically outperforms classical baselines in both error magnitude (RMSE) and directional accuracy (MDA), these results underscore a harsh reality of quantitative finance: exceeding the random-walk threshold (MDA $> 50\%$) is a necessary mathematical milestone, yet vastly insufficient for guaranteed net profitability. Operating in such an adversarial and highly frictional environment demands far more than raw predictive accuracy. Transitioning from a theoretical deep learning model to a professional-scale trading system requires the implementation of complex execution strategies to navigate transaction costs, the integration of diverse data modalities encompassing fundamental analysis, and the continuous, real-time recalibration of risk parameters to survive abrupt regime shifts.

5. Future Work

To overcome current limitations and bridge the gap toward institutional-grade quantitative forecasting, several promising avenues for future research could be explored across three fundamental pillars:

1. **Long-Range Attention Mechanisms:** A potential direction involves exploring highly efficient architectures, such as Informers or Sparse Attention variants. These could facilitate the processing of extremely long historical sequences (e.g., high-frequency intra-minute ticks) without incurring quadratic memory bottlenecks or generating the predictive hallucinations common in standard Transformers when exposed to extended noisy contexts.
2. **Fundamental and Sentiment Integration:** Recognizing the limitations of purely technical and mathematical signals, future iterations could benefit from incorporating Natural Language Processing (NLP) to extract real-time sentiment from social media streams. Such an approach would explicitly couple these metrics with the automated parsing of overarching macroeconomic and geopolitical events.
3. **Privileged Microstructural Data:** To construct models capable of overcoming market friction, it may be necessary to move beyond publicly available, aggregated OHLCV data. Consequently, future frameworks might consider ingesting high-tier institutional or exchange-level data—such as deep Limit Order Book (LOB) snap-

shots, liquidation clusters, and internal order flow—to establish a more definitive informational edge.

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A. Appendices: Supplementary Visualizations and Extended Benchmarks

This appendix provides extended visual documentation of the quantitative benchmarks discussed in the primary text. The following bar plots detail the out-of-sample performance metrics—Root Mean Squared Error (RMSE) and Mean Directional Accuracy (MDA)—across various experimental configurations and multi-modal asset evaluations.

A.1. Experiment C: Fast Context Ablation Results

Figure 5 illustrates the performance distribution across the three temporal resolutions when the architecture is provided with a deep historical context ($S = 60$) but restricted to a single-step prediction horizon ($H = 1$). This configuration significantly elevated the MDA of the Hybrid architectures above the classical baselines.

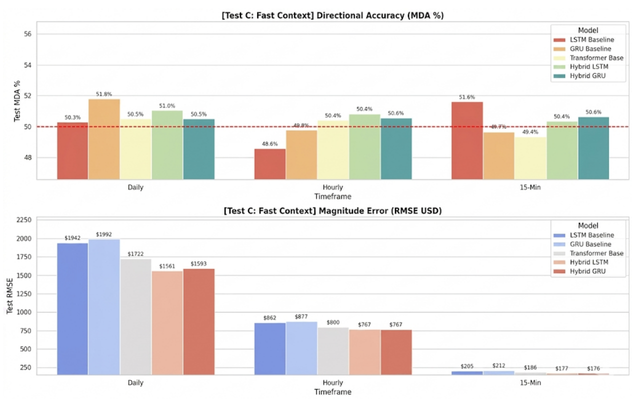

Figure 5. Performance metrics (MDA and RMSE) for Experiment C ($60 \rightarrow 1$ horizon). The Hybrid models demonstrate superior directional accuracy.

A.2. Optimized Framework: PACF Autotuning and DVA Loss

Figure 6 showcases the culmination of the optimization protocol on the Bitcoin dataset. By mathematically anchoring the sequence length to the Partial Autocorrelation Function (PACF) of market volatility and scaling the directional penalty via the real-time VIX index, the Hybrid architectures achieve their most robust error compression.

A.3. Cross-Asset Generalization: Ethereum (ETH-USD)

To try out-of-domain implementations, Figure 7 presents the evaluation of the proposed framework trained from scratch on Ethereum. The significant reduction in MDA, as well as MAE, point to a challengin scenario for altcoin price forecasting.

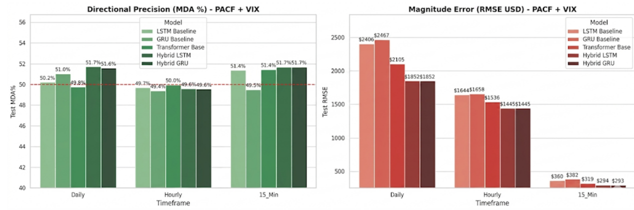

Figure 6. Final Benchmark on Bitcoin utilizing Box-Jenkins PACF context optimization and the Dynamic VIX-Adjusted Loss function.

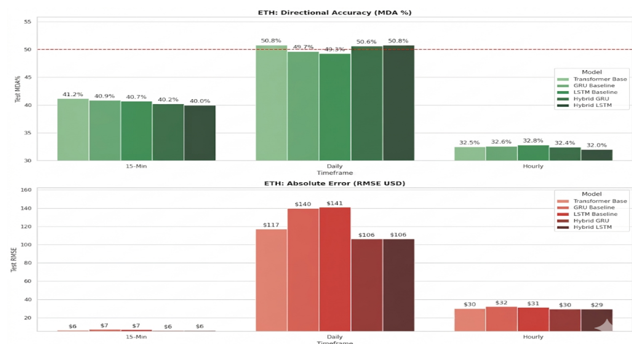

Figure 7. Generalization performance (MDA and RMSE) of the optimized hybrid framework applied to the Ethereum (ETH-USD) dataset.

A.4. Cross-Asset Generalization: Solana (SOL-USD)

Figure 8 details the framework's performance on Solana, a highly volatile alternative crypto-asset. Just as ethereum, metrics show that these assets' volatilty hinder our architecture's approach compared to Bitcoin forecasting.

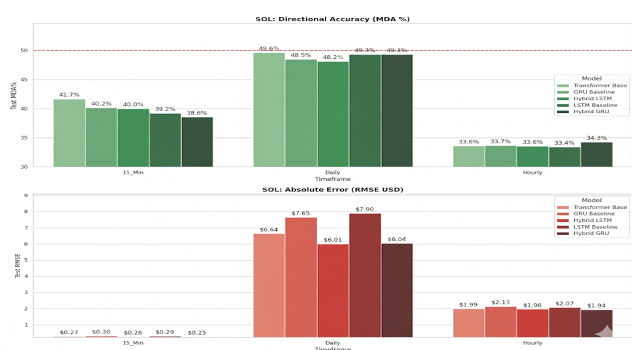

Figure 8. Generalization performance (MDA and RMSE) of the optimized hybrid framework applied to the Solana (SOL-USD) dataset.